

# Testing phylogenetic signal under rate heterogeneity and invariant-site mixtures with SatuTe

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## Abstract

Sequence saturation can leave a branch in a phylogeny without enough signal to justify the split it represents. SatuTe tests this by estimating a coherence coefficient for the second largest eigenvalue of the substitution model. We ask whether, under rate-heterogeneous Tree-of-Life models, the remaining non-stationary directions should contribute to the same test, and how an explicit invariant-site component changes the infinite-branch null. The relative eigenvalue-weighted statistic keeps the SatuTe null hypothesis and reduces to SatuTe under the Jukes–Cantor model. For  $+I$  mixtures, the invariant category remains jointly distributed across an infinite focal branch and must enter only the null denominator. The publication-scale fixed-model suite uses GTR+F, GTR+F+G4 and GTR+F+I+G4, with 1,000 paired replicates per grid cell and independent truth-labelled exact-null and mixed-null calibration. In the matched GTR+F+G4 grid, eigenvalue weighting increased the largest unadjusted informative-call fraction by 24.1 percentage points, by 57.6 points under taxon Bonferroni, and by 41.4 points under Benjamini–Yekutieli. Across 1,728 matched-protein cells the weighted difference was positive in 831, zero in 897 and negative in none. Mixed-null GTR power was 92.9–98.9%, with empirical FDR at most 0.39%. The paired effects were concentrated in short alignments and long branches, while JC remained an exact zero-difference control. The older dense  $+I + R4$  calculation is retained as an exploratory methodological analysis; all primary rate-heterogeneity claims use the fixed  $+G4$  and  $+I + G4$  specifications.

Saturation occurs when repeated substitutions erase phylogenetic information from an alignment. For two sequences, saturation means that the alignment no longer justifies their inferred evolutionary separation. In larger alignments, the same problem appears at the level of branches. A branch can appear in a maximum-likelihood tree, and can have high conventional support, while the alignment carries little signal for the two groups separated by that branch.

SatuTe generalises saturation between two sequences to a test of saturation between subtrees [1]. A branch  $AB$  splits a site pattern  $\partial$  into subpatterns  $\partial_A$  and  $\partial_B$  on the subtrees  $T_A$  and  $T_B$ . The null hypothesis is that the two subtrees evolve independently. Rejection implies phylogenetic signal for branch  $AB$ ; otherwise, the branch is considered saturated.

The test uses the spectral decomposition of the reversible substitution model [2]. For a site pattern  $\partial$ , SatuTe computes partial likelihood vectors  $L(\partial_A)$  and  $L(\partial_B)$ . It also computes subtree pattern probabilities  $P(\partial_A) = \langle L(\partial_A), \pi \rangle$  and  $P(\partial_B) = \langle L(\partial_B), \pi \rangle$ . With eigenvalues  $\lambda_0 = 0 > \lambda_1 \geq \lambda_2 \geq \dots$  and left eigenvectors  $h_i$ , the coherence coefficient for a simple eigenvalue is

$$C_{\partial i} = \left\langle \frac{L(\partial_A)}{P(\partial_A)}, h_i \right\rangle \left\langle \frac{L(\partial_B)}{P(\partial_B)}, h_i \right\rangle. \quad (1)$$

SatuTe uses this coefficient for the second largest eigenvalue. Across  $n$  sites, the empirical statistic is  $\hat{C}_1 = n^{-1} \sum_{s=1}^n C_{\partial_s 1}$ , with variance estimator  $\hat{\sigma}_1^2$ . The one-sided statistic is  $Z = \sqrt{n} \hat{C}_1 / \hat{\sigma}_1$ . The test rejects subtree independence when  $Z > z_\alpha$ .

The second largest eigenvalue gives the slowest non-stationary decay towards the stationary distribution. It is therefore the conservative component on long branches, because all other

non-stationary components decay faster. Under general time-reversible nucleotide models [3] and amino-acid models such as LG, these non-stationary eigenvalues are usually distinct. We ask whether including the faster components changes which branches or alignment regions are called phylogenetically informative when a fitted rate-heterogeneous model separates them.

## Results

### A weighted coefficient keeps the SatuTe test

The spectral decomposition used by SatuTe gives

$$\frac{P(\partial | t)}{P(\partial_A)P(\partial_B)} = 1 + \sum_{i=1}^d C_{\partial i} \exp(\lambda_i t). \quad (2)$$

Here  $d$  is the number of non-stationary eigenvalues:  $d = 3$  for DNA and  $d = 19$  for an amino-acid model. The coefficient  $C_{\partial i}$  measures agreement between the two subtrees in spectral direction  $h_i$ . The factor  $\exp(\lambda_i t)$  measures how much of that direction remains after evolution across branch  $AB$ . Because all non-stationary eigenvalues are negative, components with more negative  $\lambda_i$  decay faster.

We define the eigenvalue-weighted coherence for pattern  $\partial$  as

$$C_{\partial}^{\lambda}(t) = \sum_{i=1}^d w_i(t) C_{\partial i}, \quad w_i(t) = \exp\{(\lambda_i - \lambda_1)t\}. \quad (3)$$

The relative weight  $w_i(t)$  sets  $w_1(t) = 1$  and downweights components that decay faster than the second-largest-eigenvalue component. Thus the published SatuTe coefficient remains the reference scale, while the other components contribute according to the fitted branch length.

For an alignment of  $n$  independently evolving sites, the empirical coefficient is

$$\hat{C}_{\lambda}(t) = \frac{1}{n} \sum_{s=1}^n C_{\partial s}^{\lambda}(t). \quad (4)$$

The variance estimator has the same form as the SatuTe estimator for a repeated second largest eigenvalue, with the mode weights included:

$$\hat{\sigma}_{\lambda}^2(t) = \sum_{i=1}^d \sum_{j=1}^d w_i(t) w_j(t) \hat{\sigma}_{A,i,j}^2 \hat{\sigma}_{B,i,j}^2. \quad (5)$$

The corresponding test statistic is

$$Z_{\lambda}(t) = \frac{\sqrt{n} \hat{C}_{\lambda}(t)}{\hat{\sigma}_{\lambda}(t)}. \quad (6)$$

The decision rule is unchanged. Branch  $AB$  is called phylogenetically informative when  $Z_{\lambda}(t) > z_{\alpha}$ . When the tree is inferred from the same alignment, the same Bonferroni correction used by SatuTe gives the threshold  $z_{\alpha/(m_A m_B)}$ .

### Invariant categories require a different null likelihood

An explicit invariant category is a boundary case of the rate mixture. Let  $q_c$  and  $r_c$  be the proportion and rate of category  $c$ , and let  $A_{sc}$  and  $B_{sc}$  be the two subtree-pattern probabilities for site pattern  $s$ . For a positive-rate category, the infinite-branch null contribution is  $q_c A_{sc} B_{sc}$ . For the invariant category  $c = 0$ , however,  $r_0 = 0$  and the transition matrix remains the identity

# SatuTe in a nutshell

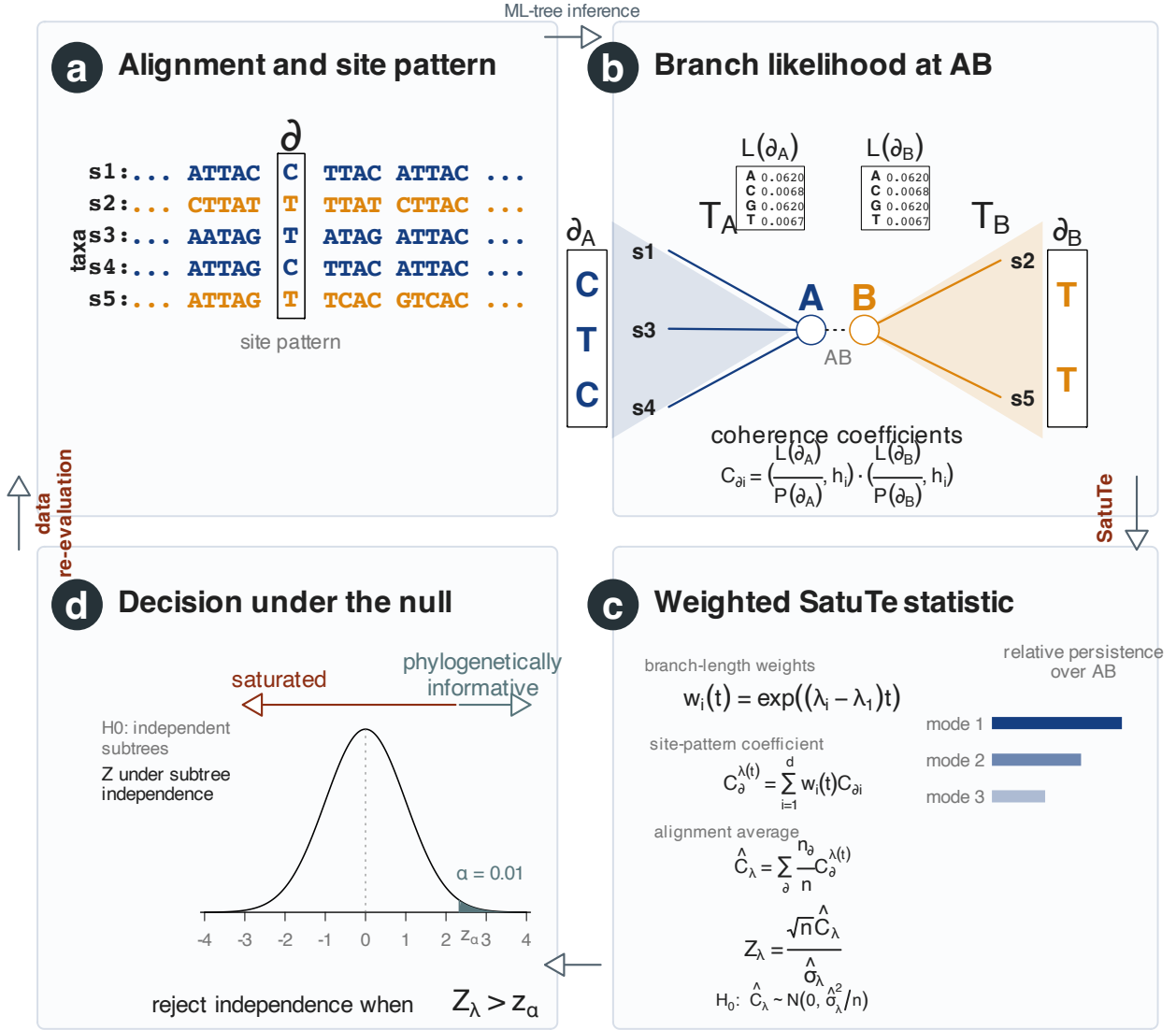


Figure 1: **SatuTe in a nutshell.** **a**, Five-taxon DNA alignment where each column represents a pattern  $\partial$ . **b**, The branch  $AB$  splits the five-taxon tree into subtree  $T_A$  with sequences  $S1, S3, S4$  and subtree  $T_B$  with sequences  $S2, S5$ . Branch  $AB$  also splits each pattern, such as  $\partial = CTTCT$ , into subpatterns  $\partial_A = CTC$  and  $\partial_B = TT$ . These subpatterns are used to compute the likelihood vectors  $L(\partial_A)$  and  $L(\partial_B)$ , which determine the coherence coefficients. **c**, The weighted statistic used here keeps the SatuTe coherence coefficients and weights each non-stationary component by its relative persistence over branch length  $t$ . **d**, The null hypothesis of independently evolving subtrees is rejected at the significance level  $\alpha$  if  $Z_\lambda > z_\alpha$ . Rejection implies phylogenetic signal for branch  $AB$ .

even as the focal branch length tends to infinity. Its null contribution is therefore the joint probability

$$J_{s0} = \sum_a \pi_a L_{A,s0}(a) L_{B,s0}(a), \quad (7)$$

not the product  $A_{s0}B_{s0}$ . The correct mixture null is

$$p_\infty(s) = q_0 J_{s0} + \sum_{c:r_c>0} q_c A_{sc} B_{sc}. \quad (8)$$

For the positive-rate categories, define the model-weighted site score

$$u_s(t) = \frac{\sum_{c:r_c>0} q_c A_{sc} B_{sc} \sum_{i=1}^d e^{\lambda_i r_c t} C_{sc,i}}{p_\infty(s)}. \quad (9)$$

The invariant component enters only the denominator and contributes no finite-versus-infinite-branch signal. This gives the exact identity

$$\frac{p_t(s)}{p_\infty(s)} = 1 + u_s(t). \quad (10)$$

In computation, all positive-rate persistence weights are divided by one common factor  $e^\delta$ , where  $\delta = \max_{c:r_c>0,i} \lambda_i r_c t$ . This prevents underflow without changing the studentized statistic. The refined test uses the sample mean and sample variance of these globally rescaled site scores. We refer to it as the refined invariant-null statistic.

The previous counterfactual construction treated the invariant category as if it became independent across an infinite focal branch, replacing  $J_{s0}$  by  $A_{s0}B_{s0}$  and retaining a zero-rate numerator contribution. That construction does not satisfy Eq. (10) under a  $+I$  model and need not have null expectation zero.

### The Jukes–Cantor case is unchanged

The Jukes–Cantor (JC) model has one stationary eigenvalue and a three-dimensional non-stationary eigenspace with  $\lambda_1 = \lambda_2 = \lambda_3$ . Therefore  $w_i(t) = 1$  for all non-stationary modes in Eq. (3). The weighted statistic becomes

$$C_\delta^\lambda(t) = \sum_{k=1}^3 C_{\partial 1,k} = C_{\partial 1}, \quad (11)$$

which is the multiplicity-three expression used by SatuTe. Thus the weighted statistic does not change the JC statistic. If the substitution model does not distinguish among non-stationary directions, the weighted statistic does not introduce that distinction.

### The original SatuTe simulations define the baseline

The simulations follow the two topologies used in the original SatuTe paper. In the five-taxon case, the tested external branch  $AB$  connects taxon  $B$  to the root  $A$  of a balanced four-taxon subtree  $T_A$ . All branches inside  $T_A$  have length 0.2, and the length of  $AB$  is varied. In the 16-taxon case,  $AB$  connects the roots  $A$  and  $B$  of two balanced eight-taxon subtrees. Branch lengths inside these subtrees are sampled from EvoNAPS-derived empirical branch-length distributions in the interval 0.04 to 0.2. Each simulated alignment is analyzed under four settings. Two settings use the true topology, with either fixed true branch lengths or maximum-likelihood branch lengths. The other two use a maximum-likelihood inferred tree, without or with the Bonferroni correction used by SatuTe.

## Rate-heterogeneous simulations use full alignments

The rate-heterogeneous comparison uses full alignments, not sliding windows. Each replicate is evaluated twice on the same tree, branch length, target split and random seed. One evaluation uses the published SatuTe coefficient  $C_{\partial 1}$ , and the other uses the eigenvalue-weighted coefficient  $C_{\partial}^{\lambda}(t)$ . The paired design isolates the effect of including the additional non-stationary components.

The first grid used 100 replicates per condition and increased the alignment length from 100 to 10,000 sites. The second grid used 1,000 replicates per condition for 100, 250 and 1,000 sites on the same five-taxon external and 16-taxon internal target branches. Both grids used Tree-of-Life-derived four-category rate-heterogeneous models. The nucleotide setting was GTR+F+G4 fitted to the 16S rRNA Tree-of-Life example, and the protein setting was LG+G4 fitted to the Tree-of-Life protein example.

## The corrected invariant null restores false-positive control

This dense exact-enumeration experiment is retained as an exploratory methodological validation of the finite-branch  $+I$  null. It is not the publication-scale rate-heterogeneity benchmark: the primary comparisons below use the fixed GTR+F+G4 and GTR+F+I+G4 models and the independent calibration suite reported in the publication-scale section. In particular, the exploratory  $+I + R4$  values below are not relabelled as  $+I + G4$  results.

The dense invariant-mixture grid evaluated 270 model-branch-alignment-length cells. With 20,000 null and 20,000 alternative replicates per cell, 10.8 million paired alignments were evaluated by both formulas. Exact enumeration verified Eq. (10) with maximum absolute error  $1.55 \times 10^{-11}$ . The refined score had exact null expectation below  $1.53 \times 10^{-16}$  in absolute value. The counterfactual score retained a positive null expectation between 0.0354 and 0.2176 across the branch grid.

At the ordinary one-sided  $N(0, 1)$  threshold, the counterfactual null rejection rate averaged 0.900 across the 270 cells and ranged from 0.500 to 1.000. The refined rejection rate averaged 0.0492 and ranged from 0.0423 to 0.0591. Splitting the null sample into threshold-training and held-out halves restored empirical rejection rates near 0.05 for both formulas, with means of 0.0501 and 0.0499, respectively. This calibration did not repair the counterfactual statistic as a standard-normal pivot: at  $t = 10,000$ , its empirical threshold increased from approximately 3.15 at 100 sites to 18.47 at 10,000 sites. The refined threshold remained between 1.62 and 1.69.

## The refined invariant statistic gains power at the detection boundary

After independent null calibration, the refined statistic had the larger exact null-centered standardized separation in 245 of 270 cells and equal separation in the remaining 25 extreme-saturation cells; it was never lower. The observed paired power difference was concentrated in a diagonal ridge: longer alignments moved the detection boundary to longer branches. The largest gain for each alignment length was:

At the largest  $+I + R4$  gain,  $t = 35$  and 10,000 sites, the paired power difference was 0.163 with bootstrap 95% interval  $[0.157, 0.170]$ . The largest  $+I + G4$  gain was 0.116 at  $t = 60$  and 10,000 sites, with interval  $[0.111, 0.122]$ . The most negative observed difference among all 270 cells was  $-0.0059$ . These sub-percentage-point losses occurred mainly where both calibrated powers were near the null floor; with 270 unadjusted cell-wise comparisons they are compatible with Monte Carlo threshold variation and were not supported by a lower exact standardized separation.

## Exploratory power curves and the publication suite

The legacy simulation curves below are retained as an exploratory overview of the JC and Tree-of-Life rate-heterogeneous settings. They are not the source of the primary publication claims: those

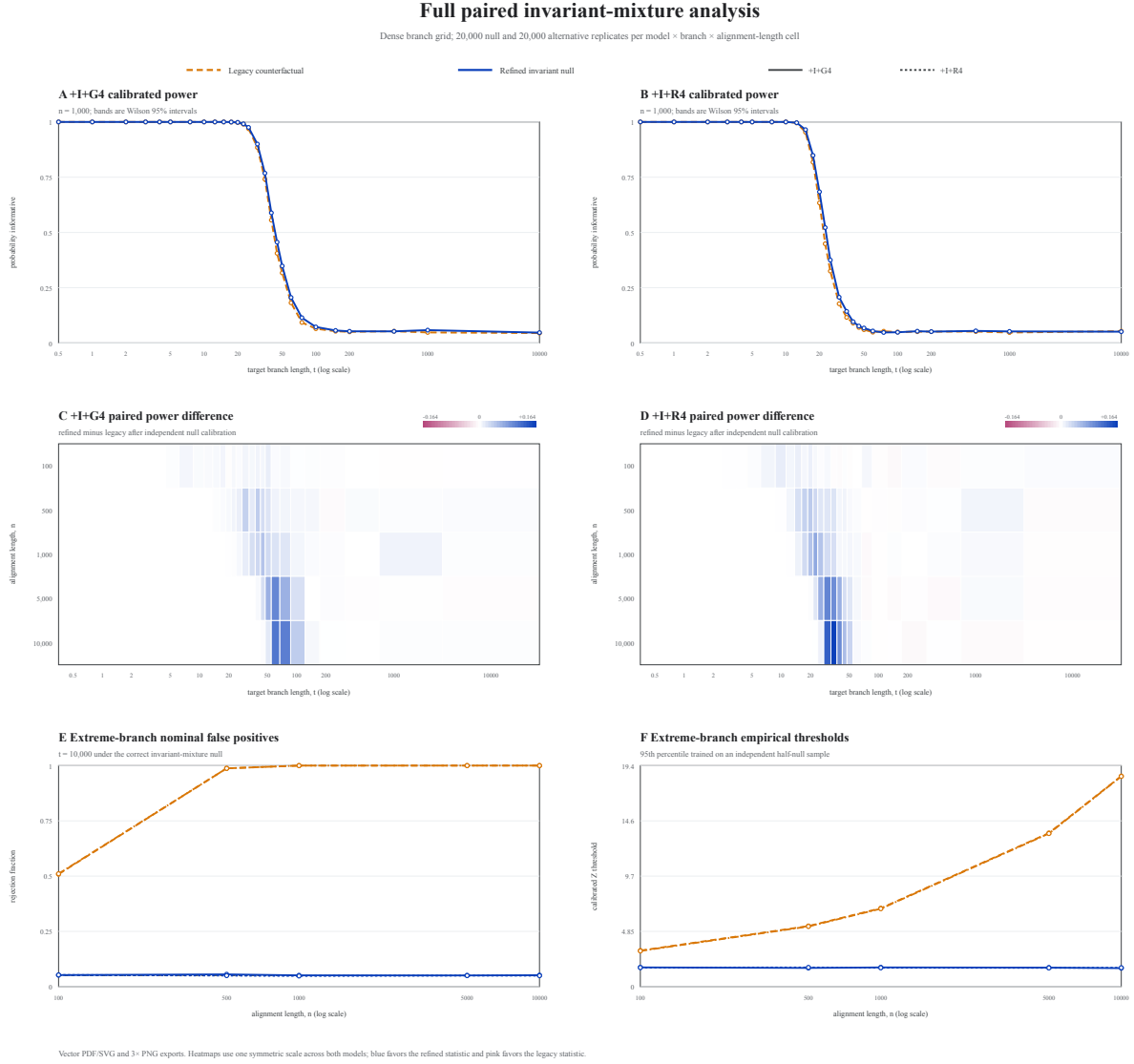


Figure 2: **Full paired +I + G4 and +I + R4 invariant-mixture analysis.** **a,b**, Formula-specific empirically calibrated power at 1,000 sites across the 27-step branch grid; bands are Wilson 95% intervals. **c,d**, Refined-minus-counterfactual calibrated power for all five alignment lengths. Blue cells favor the refined statistic and pink cells favor the counterfactual statistic; both panels share one symmetric color scale. **e**, Nominal false-positive rates at  $t = 10,000$ . **f**, Formula-specific empirical 95th-percentile thresholds trained on an independent half-null sample.

| Model        | Sites  | Branch $t$ | Counterfactual | Refined | Difference |
|--------------|--------|------------|----------------|---------|------------|
| + $I$ + $G4$ | 100    | 50         | 0.100          | 0.115   | 0.015      |
|              | 500    | 40         | 0.342          | 0.384   | 0.042      |
|              | 1,000  | 45         | 0.406          | 0.456   | 0.050      |
|              | 5,000  | 60         | 0.482          | 0.584   | 0.102      |
|              | 10,000 | 60         | 0.718          | 0.835   | 0.116      |
| + $I$ + $R4$ | 100    | 10         | 0.558          | 0.573   | 0.015      |
|              | 500    | 20         | 0.405          | 0.450   | 0.044      |
|              | 1,000  | 22.5       | 0.449          | 0.522   | 0.073      |
|              | 5,000  | 30         | 0.501          | 0.605   | 0.104      |
|              | 10,000 | 35         | 0.402          | 0.565   | 0.163      |

Table 1: **Largest calibrated power gain at each alignment length in the dense invariant-mixture grid.** Differences are refined minus counterfactual. Every alignment is evaluated by both formulas.

claims use the corrected 1,000-replicate publication grid, fixed GTR+F/G4/+I+G4 specifications, explicit model- misspecification comparisons, and independent truth-labelled calibration reported in Section 24. Under JC, the published and weighted curves coincide, providing the expected zero-difference control. The publication-scale GTR effects and their paired uncertainty intervals are reported from the verified schema-v2 tables in Section 24.

The earlier 1,000-replicate curves give the qualitative pattern that motivates the corrected suite: larger separations occur for short alignments and long branches, while decisions converge as alignment length increases. Because the legacy grid used a different cell definition and model set, its cell-wise percentages are not pooled with the publication estimates; the complete publication-scale estimates, intervals, and McNemar tests are given in Section 24.

## Decision changes are localized

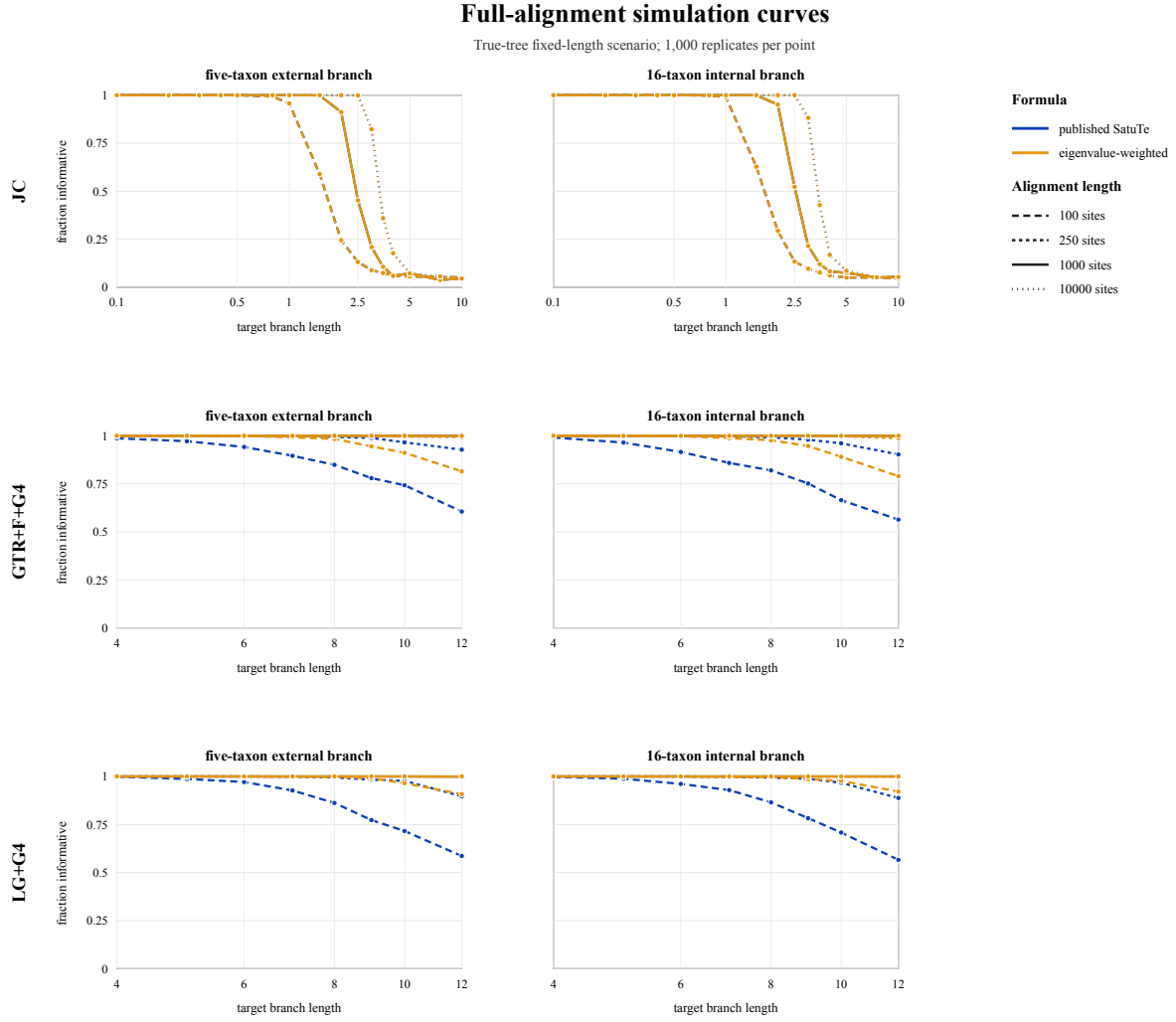
The branch-neighborhood analysis likewise remains an exploratory localization check. It tests whether changes are concentrated on the focal branch rather than its neighbours; the corrected branch-audit tables in the publication suite provide the authoritative branch-level evidence and are used for the paper’s claims.

## Biological sliding-window rerun

The biological examples use the Tree-of-Life panels from the original SatuTe paper. The protein analysis uses the ribosomal-protein 2D tree, and the nucleotide analysis uses the 16S rRNA 3D tree. In both trees, the highlighted internal split is the branch leading to Eukaryota. The yeast branch is an external branch within the Eukaryota subtree and is the second focal branch in the sliding-window analyses.

The curated biological rerun applies the same comparison to the 16S rRNA Tree-of-Life dataset shown in the 3D tree panel. The dataset has 1,947 alignment sites, a fitted GTR+F+G4 model, four rate categories and 36-site sliding windows. The comparison includes the published SatuTe rerun, a native SatuTe recomputation using  $C_{\partial 1}$ , and a native eigenvalue-weighted recomputation. Differences between the published and native SatuTe curves reflect differences in implementation and input files. The formula comparison is therefore the native SatuTe recomputation versus the native eigenvalue-weighted recomputation.

For the branch leading to Eukaryota, the published SatuTe rerun marks 822 of 1,912 windows as saturated. The native SatuTe recomputation marks 895 windows, whereas the native eigenvalue-weighted recomputation marks 699 windows. For the yeast branch, the corresponding



JC panels use 100, 1,000 and 10,000 sites; GTR+F+G4 and LG+G4 panels use 100, 250 and 1,000 sites.

**Figure 3: Exploratory full-alignment simulation curves.** Curves show the fraction of replicates called phylogenetically informative in the true-tree fixed-length scenario. These legacy curves provide context for the corrected publication-scale analyses in Section 24; under JC, the published SatuTe and eigenvalue-weighted curves overlap.



counts are 70, 83 and 29 saturated windows. The published-to-native difference is not used as a formula-effect estimate.

A second biological rerun uses an independent implementation that does not read SatuTe component files or SatuTe result trees. It recomputes the sliding-window scores from the IQ-TREE treefile, model report, site-probability output and original alignments for both Tree-of-Life examples [4]. The z-score curves show the same branch order and window size as the published sliding-window analysis. Because this rerun contrasts published outputs with an independent recomputation, the curves are not used as the primary formula-effect estimate.

In the protein 2D tree, the branch leading to Eukaryota has 2,561 windows. The saturated-window counts are 138 in the published SatuTe rerun and 5 in the native eigenvalue-weighted recomputation. The yeast branch also has 2,561 windows, with counts 54 and 8. In the 16S rRNA 3D tree, the branch leading to Eukaryota has 1,912 windows, with counts 822 and 699. The yeast branch has 1,912 windows, with counts 70 and 29. These counts summarize the threshold crossings in the sliding-window curves.

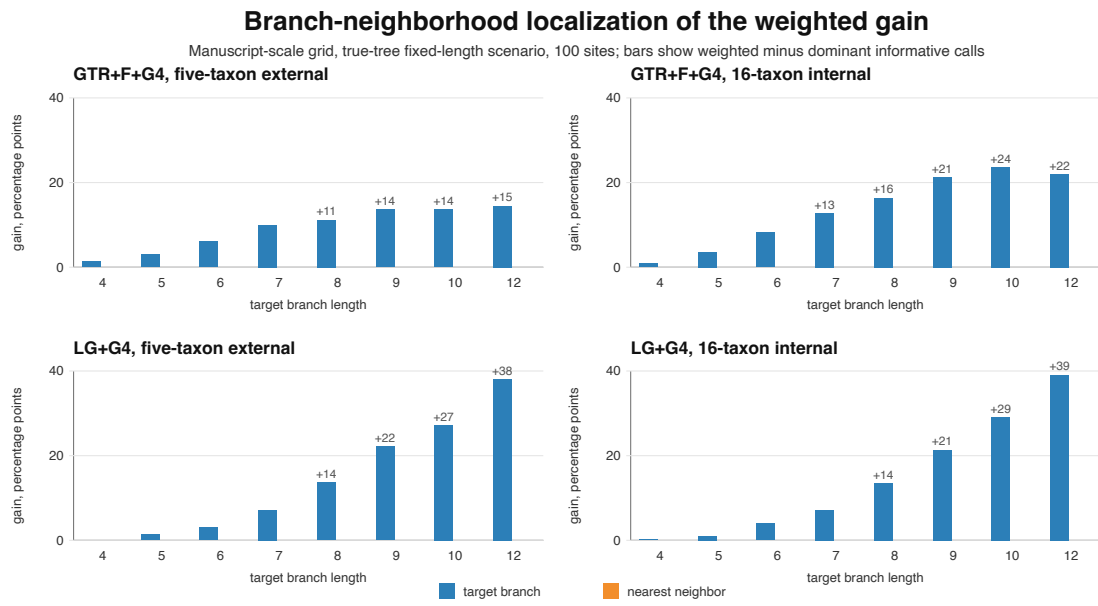
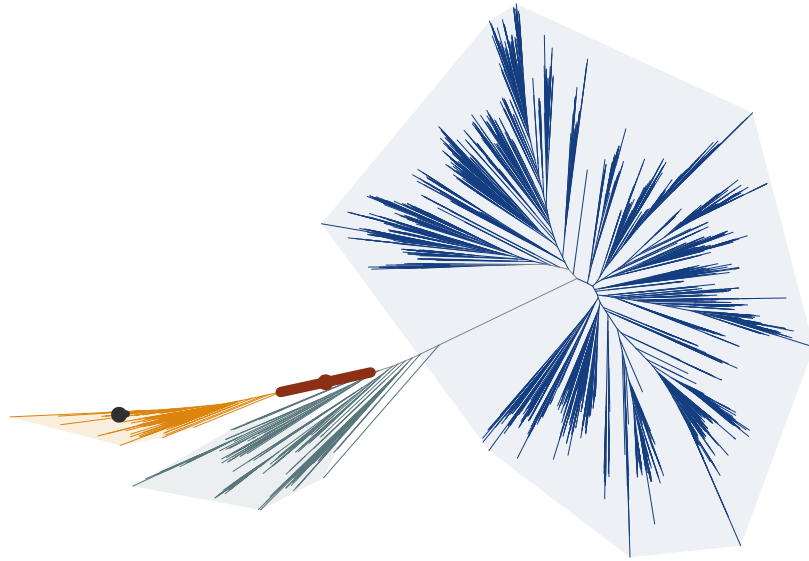


Figure 4: **Exploratory branch-neighborhood localization.** Bars show the change in informative-call fraction from eigenvalue weighting in the legacy rate-heterogeneous grid. The corrected branch-audit tables in Section 24 are the authoritative publication-scale analysis.

**(a) 2D Tree of Life: ribosomal protein alignment**



**(b) 3D Tree of Life: 16S rRNA alignment**

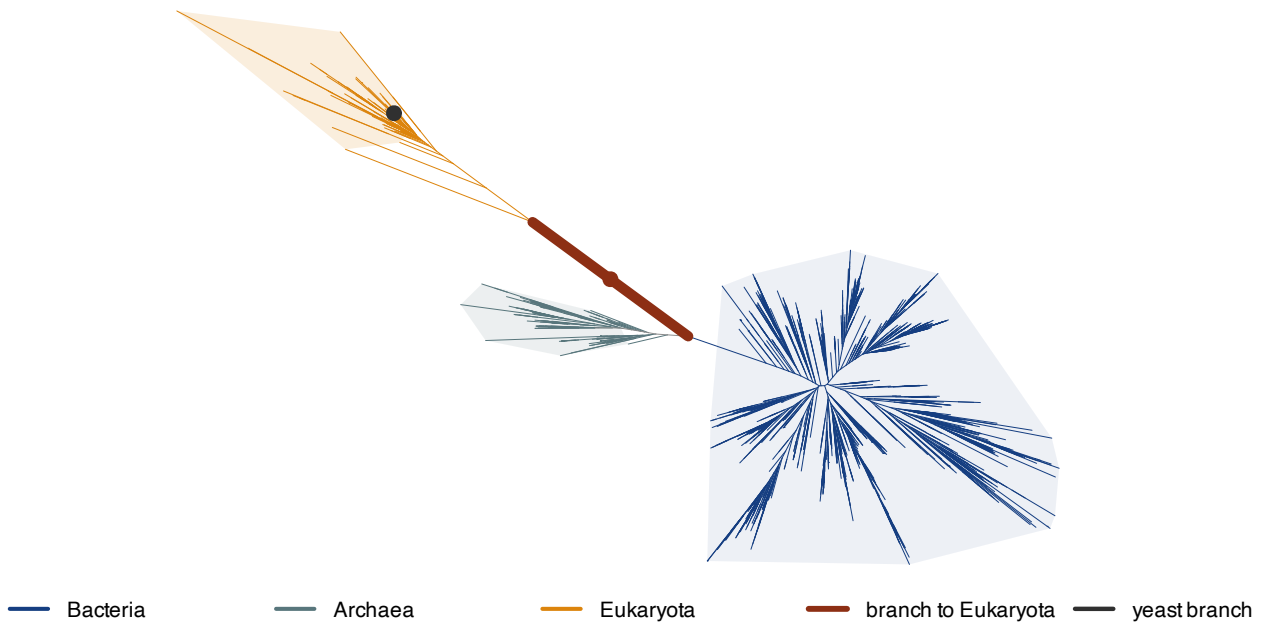


Figure 5: **Tree-of-Life references for the biological reruns.** **a**, Ribosomal-protein 2D tree. **b**, 16S rRNA 3D tree. Branch colors and translucent hulls identify Bacteria, Archaea and Eukaryota. The highlighted internal segment marks the branch leading to Eukaryota, and the black marker identifies the yeast branch used as the external-branch reference.

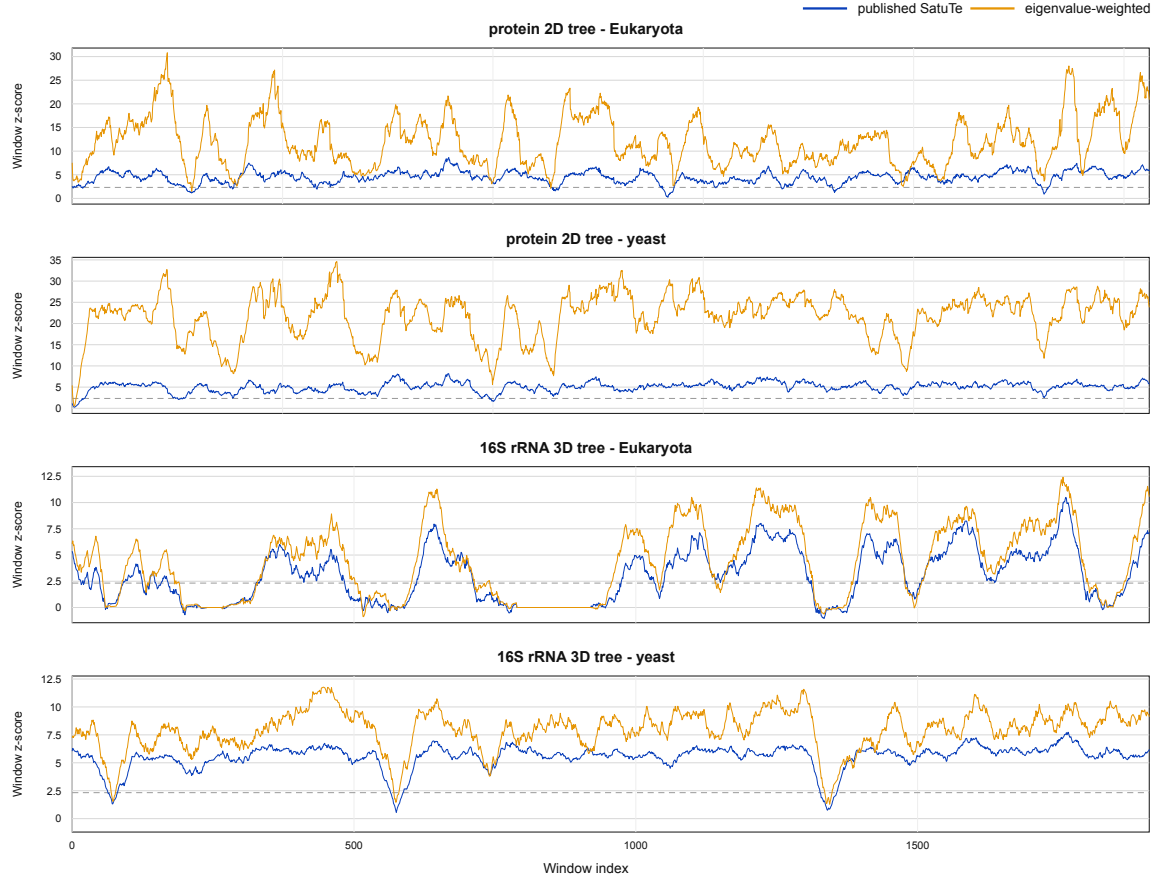


Figure 6: **Published SatuTe and independent eigenvalue-weighted sliding-window curves in the Tree-of-Life examples.** Curves show 36-site window  $Z$ -scores for the branch leading to Eukaryota and for the yeast branch in the ribosomal-protein 2D tree and the 16S rRNA 3D tree. The dashed line marks  $z_\alpha$ .

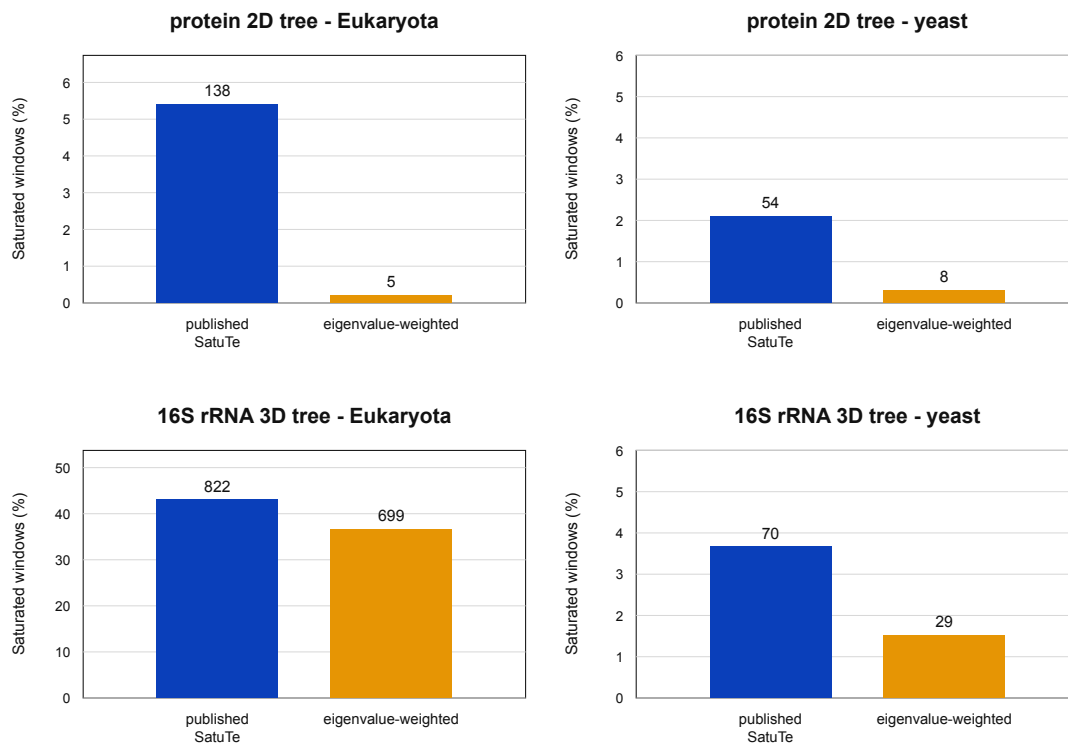


Figure 7: **Published SatuTe and independent eigenvalue-weighted sliding-window calls in the Tree-of-Life examples.** Bars show the fraction of windows classified as saturated for the branch leading to Eukaryota and for the yeast branch. Numbers above bars give saturated-window counts.

## Publication-scale GTR simulations and multiple-testing controls

We repeated the corrected head-to-head experiment on the publication grid (1,000 replicates per cell; 100, 250, and 1,000 sites; branch lengths 4–12; five-taxon external and 16-taxon internal targets) using the fixed GTR+F, GTR+F+G4 and GTR+F+I+G4 specifications. Each replicate records the dominant and eigenvalue-weighted statistics, branch-level audit, decision-rule audit, and timing information. The merged GTR grid contains 864,000 detail rows and 2,016 summary rows, with 1,000 replicates in every evaluated cell; the native calculation invariant has zero JC-collapse mismatches.

Under the unadjusted rule, the largest paired increase in the informative-call fraction was 0.241 (95% bootstrap interval 0.215–0.269; exact McNemar  $p = 5.7 \times 10^{-73}$ ) for GTR+F+G4 at 100 sites and branch length 12 on the 16-taxon target in the true-topology/ML-length scenario. With taxon-level Bonferroni correction the largest increase was 0.576 (95% interval 0.546–0.607;  $p = 8.1 \times 10^{-174}$ ), and with the Benjamini–Yekutieli rule it was 0.414 (95% interval 0.384–0.446;  $p = 4.7 \times 10^{-125}$ ). These are paired replicate comparisons, so the intervals and tests quantify the change in calls rather than treating the two formulas as independent samples.

The independent dominant, eigenvalue-weighted, and paired-difference plots are shown for the matched GTR case in Fig. 9; the fixed GTR+G4 and GTR+I+G4 paired curves are included in the supplementary figure set. The corresponding vector 3D views preserve the site, branch-length, and tree-case dimensions without rasterising the surface (supplementary figures).

We then evaluated the same simulated GTR+F alignments under JC, K2P, and F81 to quantify model misspecification. The corrected misspecification grid contains 960,000 detail rows and 2,208 summaries, with 1,000 replicates in every available cell. The largest unadjusted paired increase was 0.131 (95% interval 0.107–0.157;  $p = 2.4 \times 10^{-24}$ ) for GTR evaluated by K2P at 100 sites and branch length 8 on the 16-taxon ML tree. Under taxon Bonferroni the largest increase was 0.206 (0.176–0.236;  $p = 5.3 \times 10^{-39}$ ), and under BY it was 0.209 (0.180–0.240;  $p = 1.1 \times 10^{-38}$ ). The JC, K2P, and F81 comparisons are shown independently in the supplementary 2D and 3D figure sets.

The matched protein suite extends the same 1,000-replicate design to fixed LG+G4, WAG+G4, JTT+G4 and Q.pfam+G4 models. Across the 1,728 model–rule–grid cells, the weighted-minus-dominant informative-call difference was positive in 831 cells and zero in 897; it was negative in none. The largest unadjusted increases were 0.030 for LG (95% interval 0.020–0.041), 0.075 for WAG (0.059–0.092), 0.479 for JTT (0.447–0.510), and 0.041 for Q.pfam (0.029–0.054). Under taxon Bonferroni, the corresponding maxima were 0.327 (0.298–0.356), 0.469 (0.438–0.499), 0.555 (0.524–0.587), and 0.354 (0.325–0.384). Under Benjamini–Yekutieli they were 0.094 (0.076–0.113), 0.165 (0.142–0.188), 0.514 (0.483–0.545), and 0.118 (0.099–0.138). As in the nucleotide grid, the largest differences occurred at 100 sites and long target branches. Figure 10 shows the four model-specific taxon-Bonferroni surfaces; the unadjusted and Benjamini–Yekutieli views and all interactive 3D panels are supplied with the complete figure bundle.

The FDR calibration suite is independent of the publication curves. It uses 5,000 exact-null replicates (2,500 training and 2,500 held out) and 2,000 50:50 mixed-null replicates. It reports unadjusted, taxon-Bonferroni, and Benjamini–Yekutieli empirical FDR, power, held-out-null calibration, and paired differences. In the GTR mixed-null design, power ranged from 0.929 to 0.989 across formulas, rules, tree cases, site counts and branch lengths, whereas empirical FDR ranged from zero to 0.00386. Formula-paired power differences were between  $-0.000071$  and  $0.000571$ , and paired empirical-FDR differences were between  $-0.000071$  and  $0.000143$ . Thus the truth-labelled experiment verifies high alternative detection and low false-discovery rates for both formulas; the larger formula effects in the publication curves arise at the saturation decision boundary rather than from global miscalibration.

Training-derived exact-null thresholds were evaluated on 2,500 independent held-out nulls per cell. Across all 96 cells, held-out rejection ranged from 0.0108 to 0.0644. Unadjusted and Benjamini–Yekutieli GTR cells ranged from 0.0432 to 0.0596, while the discrete 16-taxon

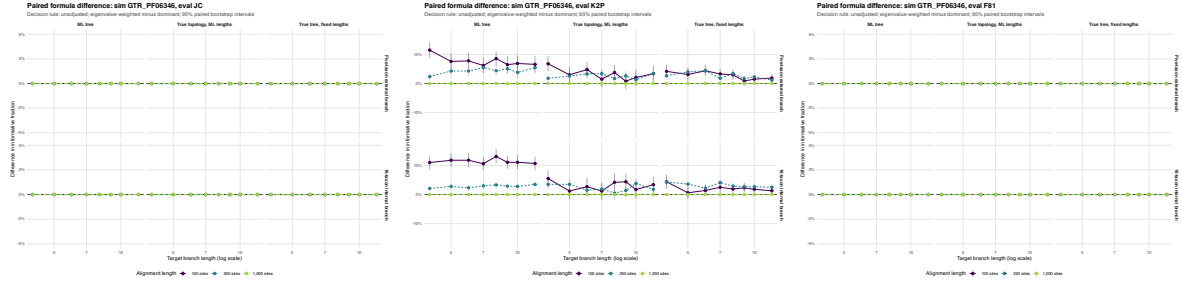


Figure 8: **GTR model-misspecification comparisons.** Paired differences between the eigenvalue-weighted and dominant formulas when the same GTR+F simulations are evaluated by JC, K2P, and F81.

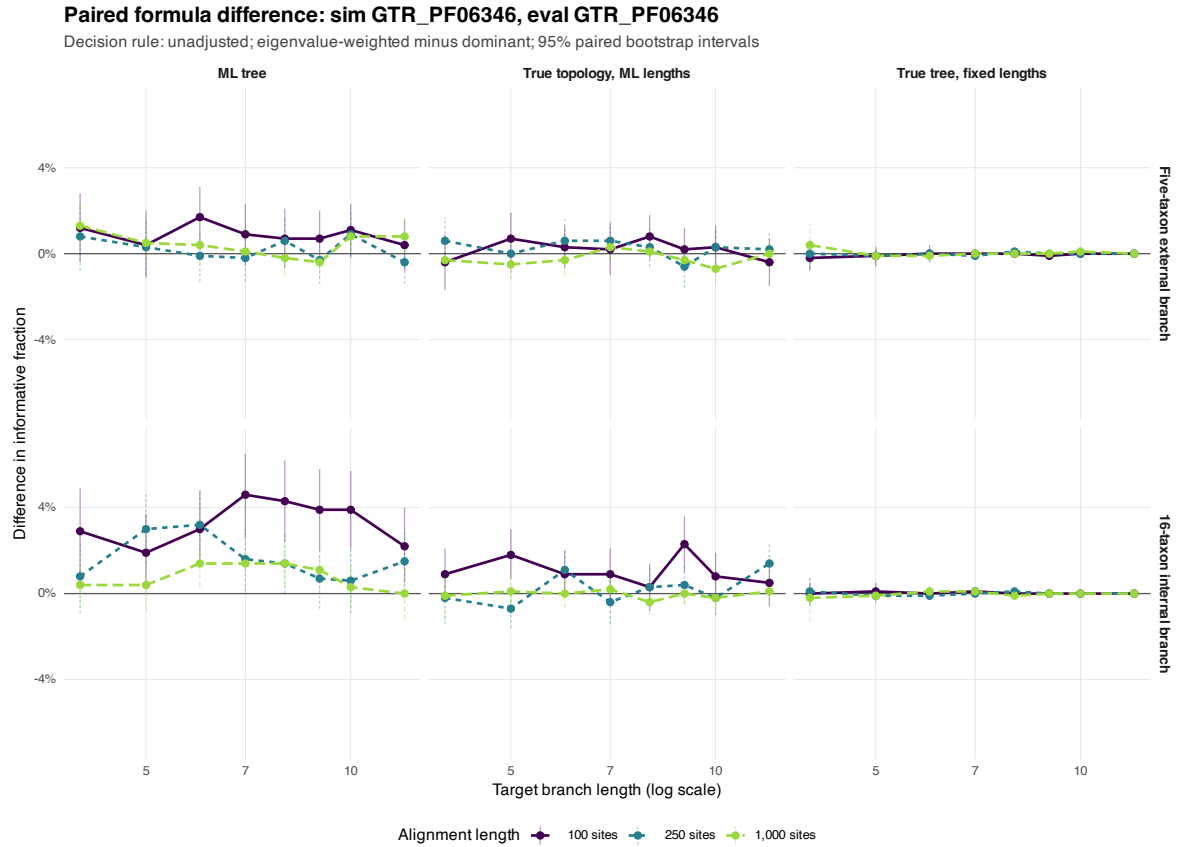


Figure 9: **Matched GTR paired difference.** Difference in informative-call fraction between the eigenvalue-weighted and published dominant formulas on the corrected 1,000-replicate grid. Curves are stratified by tree and analysis scenario; line styles encode alignment length.

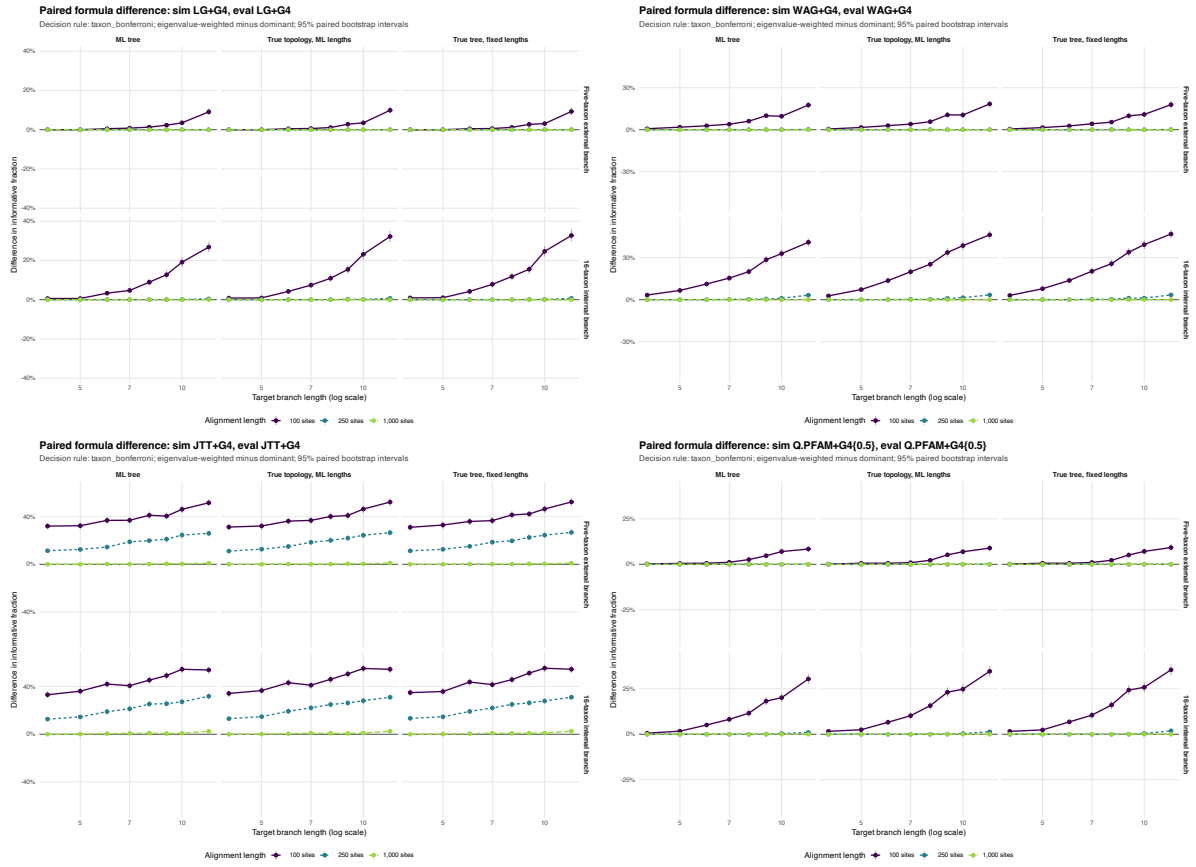


Figure 10: **Matched protein results under taxon Bonferroni.** Paired weighted-minus-dominant differences for fixed LG+G4, WAG+G4, JTT+G4 and Q.pfam+G4 models. Every panel uses the same 1,000 paired replicates per evaluated cell and reports bootstrap 95% intervals.



taxon-Bonferroni cells were conservative because only complete adjusted- $p$ -value tie blocks are attainable. The calibration figures are generated independently in 2D and vector 3D formats; the weighted GTR operating characteristics and held-out calibration are shown in Fig. 11.

## Discussion

The weighted statistic changes one step of SatuTe. The branch, subtree likelihood vectors, null hypothesis and one-sided test remain the same. When the fitted substitution model separates the other non-stationary components, the statistic includes them after scaling by their expected decay along branch  $AB$ .

The simulations identify the settings in which the formula changes the decision. Under JC, the statistic is unchanged. Under four-category rate-heterogeneous nucleotide and protein models, the weighted statistic changed decisions most often for 100-site alignments and long tested branches. With more sites, the published SatuTe statistic usually had enough information to call the branch phylogenetically informative, and the weighted statistic rarely changed the binary decision. Across all 1,728 matched-protein cells the observed change was never negative, but its size varied markedly among LG, WAG, JTT and Q.pfam. The independent truth-labelled experiment found nearly identical empirical FDR and power for the two formulas away from this transition. The evidence therefore supports a localized gain at the saturation decision boundary, not a universal increase in power under every model and branch regime.

The explicit invariant-site analysis identifies a separate issue. A zero-rate category never loses the shared ancestral state across the focal branch, so treating that category as independent in the infinite-branch null creates a positive score even when the data are generated from the correct null. Studentization then amplifies the shift at rate  $\sqrt{n}$ , explaining why the counterfactual nominal rejection rate approaches one as alignment length grows. The refined denominator restores exact null centering and ordinary standard-normal calibration. Once both formulas are given empirical thresholds, the remaining power change is smaller and localized to the detection boundary, where the refined statistic gained as much as 16.3 percentage points in the tested grid.

The dense grid also clarifies resolution. A coarse set of branch lengths captures the false-positive failure but understates the peak power difference because the detection transition moves with alignment length and rate mixture. The 2.5-unit steps from 10 to 25 and 5-unit steps from 30 to 50 resolve the diagonal power-gain ridge in the full-grid figure. Outside this ridge, both formulas usually have either near-null or near-complete power, so a formula change has little room to alter the binary decision.

The branch-neighborhood analysis addresses branch specificity. If the weighted statistic simply made all branches easier to call informative, the same change would appear on neighboring branches. In the 1,000-replicate simulations, the decision changes were concentrated on the tested branch, while nearest neighbors were already informative under both formulas. This is consistent with a branch-specific interpretation within the simulated five-taxon and 16-taxon settings.

In the Tree-of-Life + $G4$  examples, the recomputations changed sliding-window saturation calls. These examples do not by themselves quantify the formula effect, because the published and native curves also differ in input files and implementation. The paired full-alignment simulations therefore provide the more controlled test. The sliding-window results are biological examples of the same question: whether regions of an alignment retain enough phylogenetic signal for a selected branch.

The original SatuTe paper did not resolve the true phylogenetic relationship of Archaea, Bacteria and Eukaryota, and this extension does not change that scope. SatuTe tests for saturation; it does not determine which tree is correct. The weighted statistic keeps that role. It applies when a rate-heterogeneous model has separated non-stationary eigenvalues and the analysis asks whether signal outside the second-largest-eigenvalue component should contribute.

The published SatuTe coefficient remains the simpler reference statistic.

## Online Methods

### Notation

We use the notation of SatuTe. A branch  $AB$  separates subtrees  $T_A$  and  $T_B$ . A site pattern  $\partial$  induces subpatterns  $\partial_A$  and  $\partial_B$ . The partial likelihood vectors at the endpoints are  $L(\partial_A)$  and  $L(\partial_B)$ . The stationary distribution is  $\pi$ , with  $P(\partial_A) = \langle L(\partial_A), \pi \rangle$  and  $P(\partial_B) = \langle L(\partial_B), \pi \rangle$ . The reversible rate matrix has eigenvalues  $\lambda_0 = 0 > \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d$  and left eigenvectors  $h_i$ , where  $d$  is the number of non-stationary components. Coherence coefficients are defined by Eq. (1).

### Weighted estimator

For each site  $s$ ,  $C_{\partial,s,i}$  is computed for the non-stationary modes. For branch length  $t$ , the mode weight is  $w_i(t) = \exp\{(\lambda_i - \lambda_1)t\}$ . The quantities  $\hat{C}_\lambda(t)$ ,  $\hat{\sigma}_\lambda^2(t)$  and  $Z_\lambda(t)$  are then computed using Eqs. (4) to (6). If a model has a repeated second largest eigenvalue, each direction in that eigenspace has weight one. For JC, the estimator is the published SatuTe statistic.

### Null distribution

The weighted statistic keeps the SatuTe null hypothesis. Let

$$X_i(\partial_A) = \left\langle \frac{L(\partial_A)}{P(\partial_A)}, h_i \right\rangle, \quad Y_i(\partial_B) = \left\langle \frac{L(\partial_B)}{P(\partial_B)}, h_i \right\rangle.$$

Then  $C_{\partial i} = X_i Y_i$  and

$$C_\partial^\lambda(t) = \sum_{i=1}^d w_i(t) X_i Y_i.$$

Under the SatuTe null hypothesis, the two subtree patterns  $\partial_A$  and  $\partial_B$  are independent. The non-stationary eigenvectors are orthogonal to the stationary component, so the original SatuTe argument gives  $\mathbb{E}[X_i] = 0$  and  $\mathbb{E}[Y_i] = 0$ . Therefore  $\mathbb{E}[C_{\partial i}] = 0$  for every non-stationary component, and by linearity  $\mathbb{E}[C_\partial^\lambda(t)] = 0$ .

The variance follows by expanding the square of the weighted sum. Define the subtree second moments

$$\sigma_{A,i,j}^2 = \mathbb{E}[X_i X_j], \quad \sigma_{B,i,j}^2 = \mathbb{E}[Y_i Y_j].$$

Using independence of the two subtrees under the null,

$$\text{Var}\left(C_\partial^\lambda(t)\right) = \sum_{i=1}^d \sum_{j=1}^d w_i(t) w_j(t) \sigma_{A,i,j}^2 \sigma_{B,i,j}^2. \quad (12)$$

For  $n$  independent sites,  $\text{Var}[\hat{C}_\lambda(t)] = \text{Var}(C_\partial^\lambda(t))/n$ , which is the population form estimated in Eq. (5). With finite fourth moments and a consistent variance estimator, the central limit theorem and Slutsky's theorem give  $Z_\lambda(t) \xrightarrow{d} N(0, 1)$  under subtree independence. Maximum-likelihood topology selection remains a separate model-selection step, so the same Bonferroni correction used by SatuTe is retained for that scenario.

## Simulation protocol

The rate-heterogeneous simulations use full alignments and paired formula evaluation. The five-taxon tree is a balanced four-taxon subtree  $T_A$  with all subtree branches set to 0.2 and a single taxon  $B$  attached by branch  $AB$ . The 16-taxon tree is built from two balanced eight-taxon subtrees. Each alignment is analyzed under four scenarios. Two scenarios use the true topology: one with fixed true branch lengths and one with maximum-likelihood branch lengths. The other two use the maximum-likelihood inferred tree, without or with the Bonferroni threshold  $\alpha/(m_A m_B)$ . Within each replicate, both formulas are evaluated on the same tree, branch length and target split.

The nucleotide simulations use the Tree-of-Life 16S rRNA GTR+F+G4 setting, implemented as the corresponding fixed four-category rate model used by the IQ-TREE integration [4]. The exploratory protein simulations use the Tree-of-Life LG+G4 setting, and the publication suite adds fixed WAG+G4, JTT+G4 and Q.pfam+G4 models. The growing-site grid uses 100, 250, 500, 1,000, 2,000, 4,000 and 10,000 sites. It uses target branch lengths 4, 5, 6, 7.5, 10 and 12, with 100 replicates per condition. The 1,000-replicate grid uses 100, 250 and 1,000 sites, target branch lengths 4, 5, 6, 7, 8, 9, 10 and 12. The branch-neighborhood analysis computes the same dominant and weighted statistics for the target branch and adjacent branches in the same simulated trees.

The independent calibration experiment uses 5,000 exact-null replicates in every cell, pre-specifying replicates 1–2,500 for threshold training and 2,501–5,000 for held-out evaluation. A separate 2,000-replicate design alternates target-null and target-alternative branches to form a 50:50 mixed-null sample for empirical false-discovery rate and power. For discrete adjusted  $p$ -values, the training-derived cutoff is the largest attainable value for which the complete tied block has empirical probability at most 0.05; if the minimum value already exceeds that mass, the cutoff is placed immediately below it. This tie-aware definition makes the fitted cutoff agree with the implemented  $p \leq c$  decision rule. We report Wilson intervals for held-out null rejection and paired bootstrap intervals for differences between formulas.

## Paired invariant-mixture grid

The  $+I$  analysis uses a four-taxon tree with two two-tip subtrees, terminal branches of length 0.05 and a varied focal branch. The substitution process is fixed to GTR with exchangeabilities  $\{1, 2, 1, 1, 2, 1\}$  and stationary frequencies  $\{0.30, 0.20, 0.20, 0.30\}$ . Both models have invariant proportion  $q_0 = 0.10$ . The  $+I + G4$  model uses four equal-probability discrete-gamma categories with shape 0.5. The  $+I + R4$  model uses positive-category proportions  $\{0.35, 0.25, 0.23, 0.17\}$  and rates  $\{0.0747656, 0.420849, 1.15850, 3.54214\}$ , rescaled after addition of the invariant component so that the overall mean rate is one.

The  $+I + R4$  construction is retained only as an exploratory methodological control. The publication benchmark uses the fixed  $+I + G4$  specification and reports fixed-model GTR+F+G4 and GTR+F+I+G4 comparisons separately.

All  $4^4 = 256$  site-pattern probabilities are enumerated exactly under the correct finite-branch distribution and under the infinite-branch null in Eq. (8). Each simulated alignment is a multinomial draw from one of these exact distributions. The same pattern counts are evaluated by the counterfactual and refined statistics, producing paired decisions without simulator-stream variation.

Alignment lengths are

$$n \in \{100, 500, 1000, 5000, 10000\}. \quad (13)$$

The dense focal-branch grid is

$$t \in \{0.5, 1, 2, 3, 4, 5, 7.5, 10, 12.5, 15, 17.5, 20, 22.5, 25, 30, 35, 40, 45, 50, 60, 75, 100, 150, 200, 500, 1000, 10000\}. \quad (14)$$

Thus the grid uses 2.5-unit resolution from 10 to 25 and 5-unit resolution from 30 to 50, where the detection transitions occur, while retaining short and extreme controls. For every model, branch length and alignment length, 20,000 null and 20,000 finite-branch alternative alignments are drawn. Half of the null replicates train a formula-specific empirical 95th-percentile threshold; the other half evaluates held-out false-positive control. Power is measured on the independent alternative sample. Wilson intervals quantify individual rejection fractions. Paired percentile-bootstrap intervals use 4,000 resamples of the three discordance outcomes: refined only, agreement and counterfactual only.

## Biological sliding-window reruns

The biological reruns use the same 36-site window size as the original SatuTe Tree-of-Life analysis. The curated rRNA rerun compares the published SatuTe result, a native SatuTe recomputation and a native eigenvalue-weighted recomputation for the Eukaryota and yeast branches in the 16S rRNA 3D tree. The independent rerun recomputes scores from the IQ-TREE treefile, model report, site-probability output and original alignments for both the ribosomal-protein 2D tree and the 16S rRNA 3D tree. Saturated windows are windows with  $Z \leq z_\alpha$ , following the SatuTe decision rule used for sliding-window analyses.

## Data availability

The repository contains reviewed summaries under `artifacts/releases/`, including the dense invariant-mixture release and curated biological sliding-window evidence. Raw LiSC shards and workstation runs are stored under ignored `artifacts/cluster/` and `artifacts/local/` directories. The invariant-mixture analysis is reproduced by `experiments/003_invariant_mixture/run.py`.

## Code availability

The IQ-TREE integration, plotting scripts and manuscript sources are in the repository. The repository also contains the scripts used to run the paired full-alignment and exact-pattern invariant-mixture simulations and to produce the merged summaries, high-resolution figures and branch-neighborhood outputs.

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This work builds on the SatuTe framework of Manuel and colleagues and on the IQ-TREE integration in this repository.

## Author contributions

B.S. developed the weighted-statistic formulation and prepared the manuscript.

## Competing interests

The author declares no competing interests.

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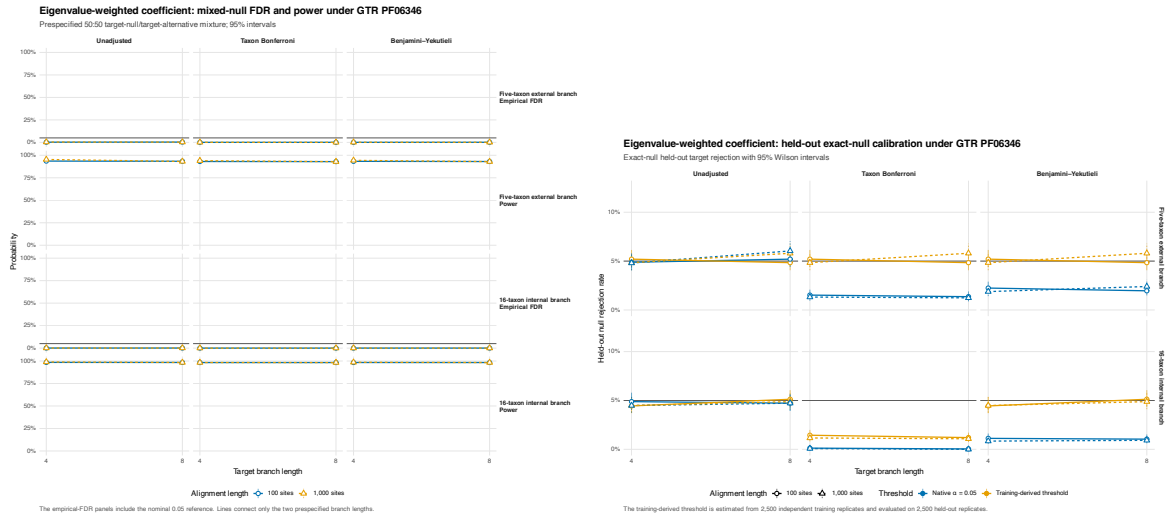


Figure 11: **Truth-labelled GTR calibration.** **Left**, empirical FDR and power of the eigenvalue-weighted formula in the prespecified 50:50 mixed-null design. **Right**, native and training-derived exact-null rejection on 2,500 held-out replicates per cell. Columns show the unadjusted, taxon-Bonferroni and Benjamini-Yekutieli rules.